Harvesting Logistic Populations

In this lab, we will consider the effects of populations of fish that grow logistically and are harvested for food. We will see that for a given population, there is *critical* harvesting rate that determines the long-term sustainability of the population. This is called the *Maximum Sustainable Yield*.

Recall the **logistic equation**

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{L}\right),\ P(0) = P_0$$

Its solution is the **the logistic function:**

$$P(t) = \frac{L}{1 + Ae^{-kt}} \qquad \text{with} \qquad A = \frac{L - P_0}{P_0}.$$

Review

1. (a) Draw a graph of $\frac{dP}{dt}$ vs. P for the logistic equation. Note that this is *not* the graph of the solution. That's P vs t!

   (b) Label the roots of your graph. What feature of the differential equation do your roots correspond to?

   (c) What term do we use for the constant L in terms of the population?

   (d) For what value of P does $\frac{dP}{dt}$ reach a maximum? What is $\frac{dP}{dt}$ at this population? Add this point to your graph and label it with its coordinates. What feature of the graph of the solution $P(t)$ does this maximum correspond to?

   (e) Sketch a rough slopefield for the logistic equation for positive time and population. How does the graph you drew above help you decide the direction of your ticks?

Harvesting

By *harvesting*, we mean removing a *fixed* number of fish from the popultion per unit time. Note that this is distinct from removing a *proportion* of the population. We deal with the latter case in class and focus on the former here.

2. First, a thought experiment:

   (a) Suppose first that the fish population is large (near its carrying capacity). If a relatively small number of fish were harvested, what effect would you expect to see on the fish population?

   (b) Next, suppose the fish population is very small. What would harvesting fish do to the population?

3. Now back to the math: suppose that a constant number F fish per unit time are removed from the population.

   (a) Write a new differential equation for this situation. You should only need to make a small change to the logistic equation.

   (b) Draw a graph of $\frac{dP}{dt}$ vs. P for this new equation. You may assume that F is small. Be sure to label L on the horizontal axis, and indicate the coordinates of the maximum value of $\frac{dP}{dt}$ on your graph. Label the roots of your graph P_1 and P_2 (with $P_1 < P_2$).

   (c) Sketch a slopefield for this equation. Use it to determine the stability of your two equilibria.

   (d) Sketch graphs of solutions ($P(t)$ vs t) for

      i. $P(0) > P_2$

      ii. $P_2 > P(0) > P_1$

      iii. $P_1 > P(0)$

   (e) Describe how your answers to the previous part correspond to your thought experiment above.

Sustainable Harvesting

We saw in the previous part that is we harvest a small number of fish, and the initial population is sufficiently high, the fish population is stable over time. We now turn to quantifying just how many fish can be harvested to maintain this stability, and understanding the consequences if this limit is exceeded.

4. (a) By examining your graphs in questions **1(d)** and **3(b)**, determine the maximal value of F that allows the population to increase. This is the *maximal sustainable yield* (MSY) of the fishery. We denote this value F_{MSY}. Note that F_{MSY} depends on both k and L!

   (b) Suppose that $F = F_{\text{MSY}}$. That is, suppose that we harvest the maximum sustainable yield. How many equilibria does the equation have in this case? Sketch a slopefield and solutions.

   (c) Describe what happens to the population of fish in the fishery if we start with a large population of fish.

5. (a) In theory, with a sufficiently large population, harvesting F_{MSY} fish per time period appears to be sustainable, as our model does not predict a collapse of the fishery. Why is this a bad idea in practice? Why might harvesting exactly the MSY of fish lead to collapse in practice?

   (b) Understanding this, the local government proposes to allow harvesting of just a small amount less than the MSY of fish. By sketching a slopefield and solutions, give arguments for and against this policy.

Unsustainable Harvesting and Collapse

6. Suppose that the harvesting rate is greater than F_{MSY}. We refer to this as *overfishing*.
   (a) Draw a graph of $\frac{dP}{dt}$ vs. P for this situation.
   (b) Are there any equilibria? What about the graph tells you that solutions still have an inflection point?
   (c) Draw a solution curve P vs. t. Be careful to indicate the inflection point.
   (d) Describe in words the change in fish population under conditions of such overfishing. Carefully refer to changing rates of population decrease in your answer.

7. Even in conditions of overfishing, it may be possible to intervene to prevent total collapse of a fishery.
   (a) By comparing the shape of your graph before the inflection point to the graphs in the sustainable cases above, explain why it might be very difficult to detect whether a population is being overfished or not merely by examining fish population levels without knowledge of the value of F_{MSY}.
   (b) Explain why, arguably, once the population is below the inflection point, the situation is dire and critical, and action should be taken immediately.
   (c) Suppose that the local government decides to intervene to mandate lower fishing levels. For this question, you will need to consider the graphs and slopefield in Question 3.
      i. Suppose that at the time of intervention, the fish population is still above the IP. Explain why, at least in theory, if fishing is reduced below F_{MSY} by any amount, the fish population is likely to recover.
      ii. Suppose by contrast that the fish population is below the IP. Explain why in this case, the amount below F_{MSY} by which fishing is reduced may determine whether or not the population will recover or still die out.
      iii. In the latter case, the government argues that it is necessary to completely stop fishing to ensure the recovery and long-term sustainability of the population. Workers who rely on fishing for a living argue that such drastic measures are not necessary. Give mathematical justifications for both arguments.

Report: The Tragedy of the Commons

In an area in which fishing has been the leading source of income, fishing families each fish only as much as they need to sustain themselves. Over time, growing local population causes fishing higher than sustainable levels for multiple common fish species, leading academics to forecast a coming collapse of the fisheries. Research and explain possible solutions. Compare and critique your proposals.

Your report should up to eight pages, and should rely on substantial and well-cited research. You should read and cite literature on sustainable fishing and the tragedy of the commons. You may use the analysis developed above as needed.